MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GENERATOR, FUNCTION, MSB, ANALOG, KEY CODE SHY

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic MSB function generator module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Name	Generator, function, MSB
Family	Analog
Key code	SHY
Size	1A
Weight	35 grams maximum
Failure rate	0.75 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Supply voltages	5.0 Vdc ±10 percent and ±15.0 Vdc ±10 percent
Maximum power dissipation	
+5.5 V (30 mA)	165 mW
+16.5 V (20 mA)	330 mW
-16.5 V (-20 mA)	330 mW
Digital inputs (internally pulled-up to V_{CC})	
2^{-n} and Q input current	
Low level (V_{IL} = 0.0 V)	-1.4 mA minimum, -0.9 mA maximum
High level (V_{IH} = 3.5 V)	-440 µA minimum, -275 µA maximum
2^{-n} and Q input voltage	
Low level (V_{IL})	0.0 V minimum, 0.8 V maximum
High level (V_{IH})	3.5 V minimum, V_{CC} maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Analog input voltage range	0.6 V minimum, 9.6 V maximum
S/D and D/S applications	
Analog input voltage	+0.6 to +9.6 V
Analog reference voltage (typical)	4.6 Vdc
Analog input current (V_I = 4.6 V)	-4.8 mA maximum
Settling time	4 us maximum
Switch dc offset voltage	1.5 mV maximum
400 and 800 Hz angular accuracy	±0.04 degrees
Phase shift	±0.5 degrees

1.3 Module function. The MSB function generator module, designated as key code SHY, is a Standard Electronic Modules Program (SEMP) electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains multipliers consisting of semiconductor switches controlled by binary digital inputs that switch resistors in or out from a precision resistor network. This module provides the necessary digital-to-analog function used in digital-to-synchro and synchro-to-digital converter applications.

1.3.1 Input to output relationship. Each MSB function generator has the input to output relationship of figure 1.

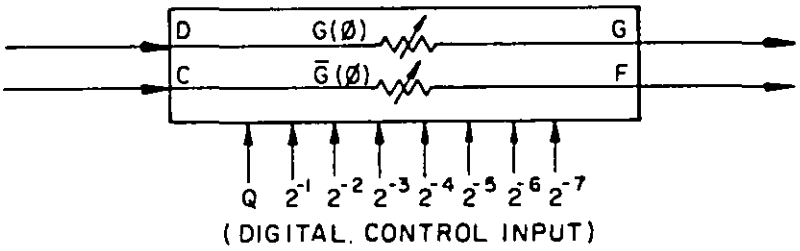


FIGURE 1. Input to output relationship block diagram.

The functions Sin θ and Cos θ are generated by the use of a nonlinear set of resistive ladder networks interconnected by the closure of solid state switching elements which in turn are actuated by the digital control inputs.

The proportionality inherent in the module is

$$G(\emptyset)_{D-G} = 0.001 \left(Q \sum_{i=1}^{7} \frac{a_i}{2^{(i-1)}} + \bar{Q} \sum_{i=1}^{7} \frac{\bar{a}_i}{2^{(i-1)}} \right) \text{ siemens (mhos)}$$

$$G(\emptyset)_{C-F} = \bar{G}(\emptyset)_{D-G}$$

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.75 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of the design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be SHY.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

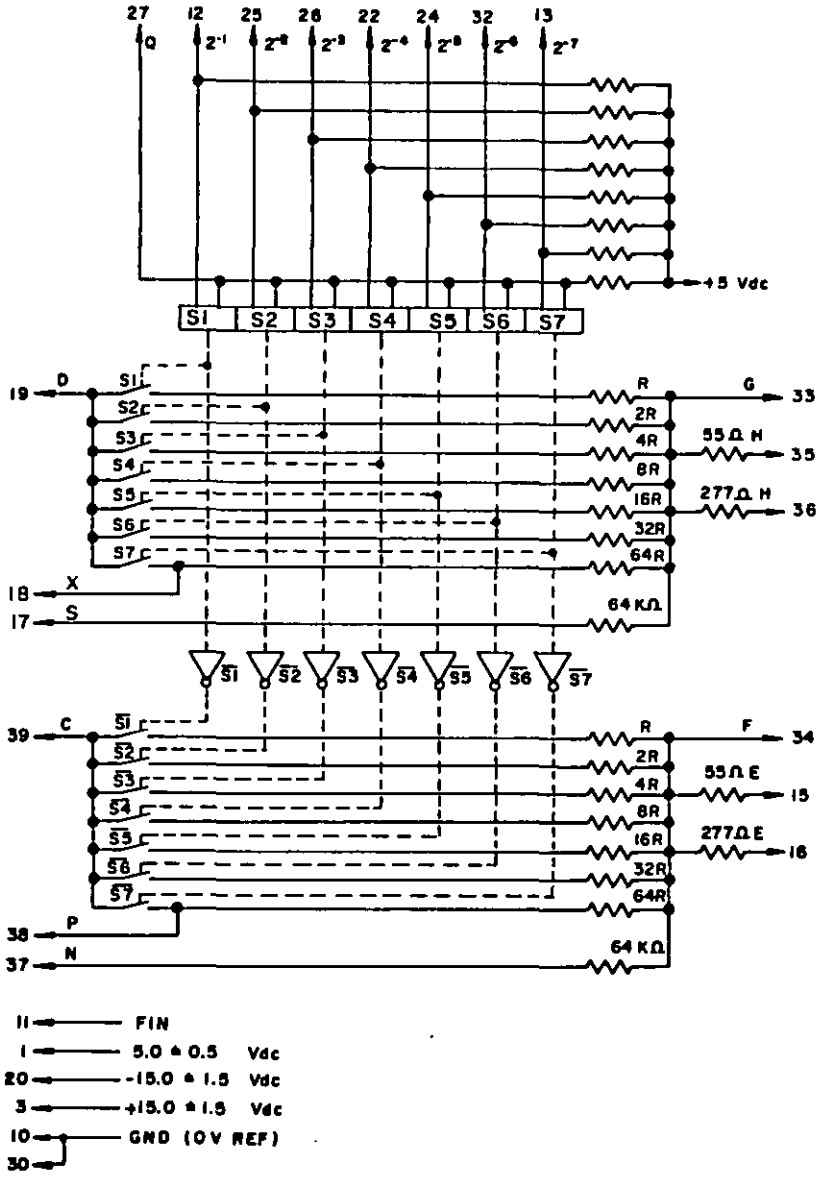
3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/186, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 2.

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NOTE: The following pins are unused: 2, 4, 5, 6, 7, 8, 9, 14, 21, 23, 28, 29, 31, and 40.

FIGURE 2. SHY module diagram, MSB function generator.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes 1-3)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A15	Isolation	II	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B1-B8	Input breakdown current	III	S	0	1	0	1	0	1	0	1	mA
C1-C6	Hot socket	IV	Q	-0.033	0.033	-0.033	0.033	-0.04	0.04	-0.04	0.04	Degrees of selected angle
D1-D14	Switch check	V	Q	← See figure 10, note 1 →								
E1, E2	Vs stress	VI	Q	-0.033	0.033	-0.033	0.033	-0.04	0.04	-0.04	0.04	Degrees of selected angle
F1-F16	Angular accuracy, 400 Hz	VII	X	-0.033	0.033	-0.033	0.033	-0.04	0.04	-0.04	0.04	Degrees of selected angle
G1-G18	DC integrity and leakage check	VIII	X	← See note 4 and table VIII →								
H1	Absolute transformation ratio check	IX	X	44.6324	44.9189	44.6324	44.9189	44.5824	44.9689	44.5824	44.9689	Degrees
H2	Phase shift	IX	X	179.5	180.5	179.5	180.5	179.5	180.5	179.5	180.5	Degrees
J1-J4	Input current (analog)	X	X	0.0	4.8	0.0	4.8	0.0	4.8	0.0	4.8	mA
K1-K8	Input current (digital), low level	XI	X	0.9	1.4	0.9	1.4	0.9	1.4	0.9	1.4	mA
K9-K16	Input current (digital), high level	XI	X	275	440	275	440	275	440	275	440	µA
L1	Power supply current, 5.5 V	XII	S	0	1	0	1	0	1	0	1	mA
L2, L3	Power supply current, ± 16.5 V	XII	S	0	20	0	20	0	20	0	20	mA
L4	Power supply current, 5.5 V	XII	Q	0	30	0	30	0	30	0	30	mA
M1-M14	Scaling resistor network	XIII	X	← See table XIII →								
N1	Continuity, ground	XIV	Q	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
N2	Continuity, ground	XIV	X	0	1	0	1	0	1	0	1	Ω
N3	Continuity, fin	XIV	S	0	1	0	1	0	1	0	1	Ω
P1-P4	Settling time	XV	X	0	4	0	4	0	4	0	4	µs

See notes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes 1-3)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q1-Q3	Angular accuracy, 800 Hz	XVI	Q	-0.033	0.033	-0.033	0.033	-0.04	0.04	-0.04	0.04	Degrees of selected angle
R1-R8	Input capacitance	XVII	Q	2	35	2	35	2	35	2	35	pF
R9-R10	Output capacitance	XVII	Q	50	100	50	100	50	100	50	100	pF
R11-R14	Output capacitance	XVII	Q	50	100	50	100	50	100	50	100	pF
S1-S6	Power supply noise rejection	XVIII	Q	0	2	0	2	0	2	0	2	mVrms
S7-S12	Power supply noise rejection	XVIII	Q	0	6	0	6	0	6	0	6	mVrms

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.
4. The initial and end-of-life limits for 25°C and operating temperature are the same.

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3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I.

3.9 Power supply voltages. The module shall be capable of conforming to the requirements of table I for power supply voltages of 5.0 ±0.5 Vdc and ±15.0 ±1.5 Vdc when driving the specified loads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
Wave analyzer	Item B03	1
Capacitance meter	Item C0D	1
Oscilloscope	Item D03	1
Synchro/resolver bridge	Item F04	1
Sine wave generator	Item G03	1
Pulse generator	Item J01	1
Digital multifunction meter	Item M02	1
DC power supply	Item P07	6
Ohmmeter	Item R01	1
Ohmmeter	Item R02	1
Megohm bridge	Item R06	1
Counter-timer	Item T01	1
DC ammeter	Item X01	1
Oscilloscope probe	Item Y04	2
Phase angle voltmeter	Item Z02	2
Resistors		
5Ω ±5%, 1/2 W		1
50Ω ±5%, 1 W		2
100Ω ±5%, 1/8		3
500Ω ±0.01%, 1/8 W		1
2 - 500Ω ±0.01% (matched to ±0.002%), 1/8 W		1 set
600Ω ±5%, 1 W		1
2 - 1 kΩ ±1% (matched to ±0.005%), 1/8 W		1 set
2.2 kΩ ±5%, 1/8 W		2
2 - 64 kΩ ±1% (matched to ±0.002%), 1/8 W		1 set
64.9 kΩ ±1%, 1/8 W		1
750Ω ±1%, 1/8 W		1
1 kΩ ±1%, 1/8 W		1
50Ω ±5%, 2 W		1

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Capacitors	No. required (maximum per test figure)
100 uF ± 5%, 20 WVdc, electrolytic	3
220 pF ± 5%, 20 WVdc	1
Diodes	
1N4148, or equivalent	2
1N4150, or equivalent	8
Miscellaneous	
OpAmp uA741, or equivalent	3
Binary counter 9316, or equivalent	2
OpAmp AD233K, or equivalent	1
Switch SPDT	3

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. *The module shall be tested as specified in table I.*

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through S shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through S in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ± 5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions F through S shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table I which are marked with an (X) in the test type column shall be performed using the initial limits at 25° ± 5°C ambient.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test, electrical tests F through S shall be performed in the order specified in 4.4.2 using the 25° ± 5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

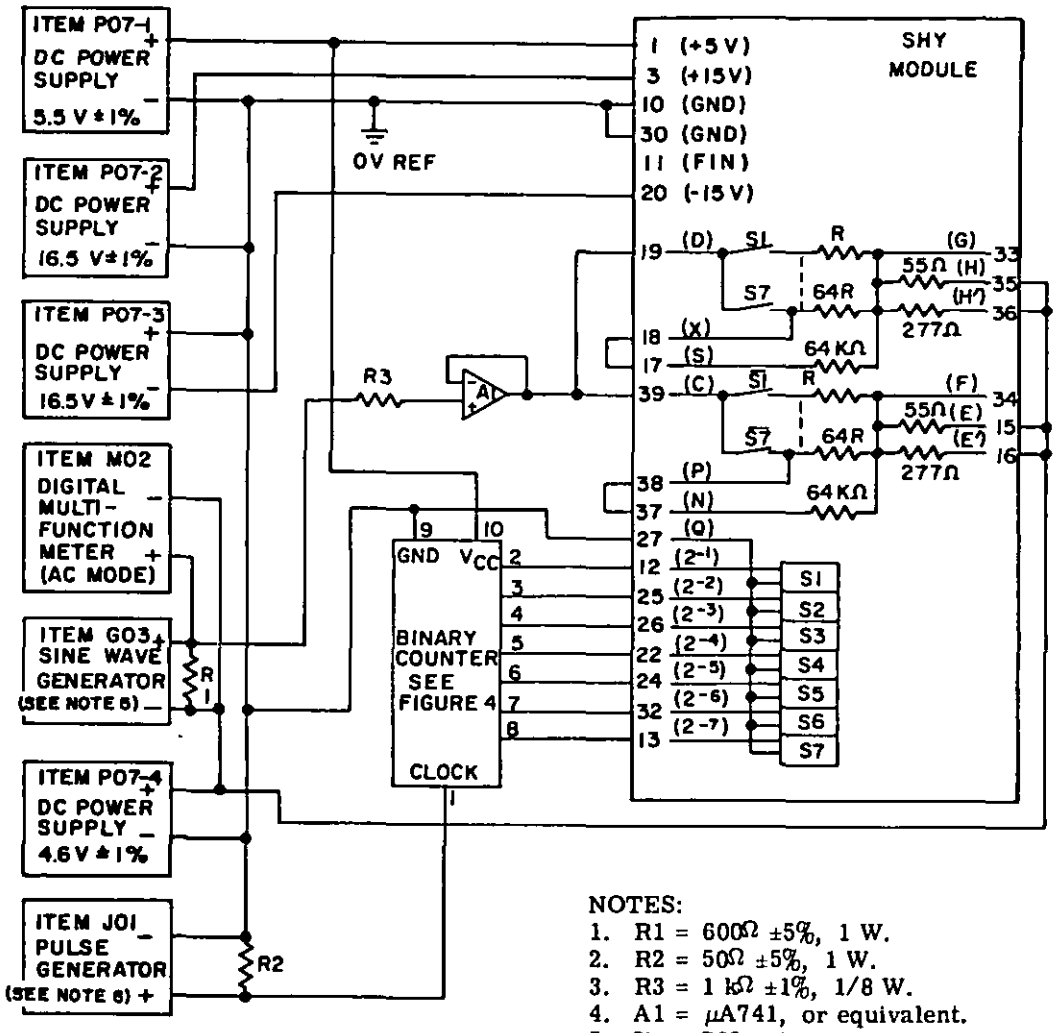
6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project MISC-NC71-3)

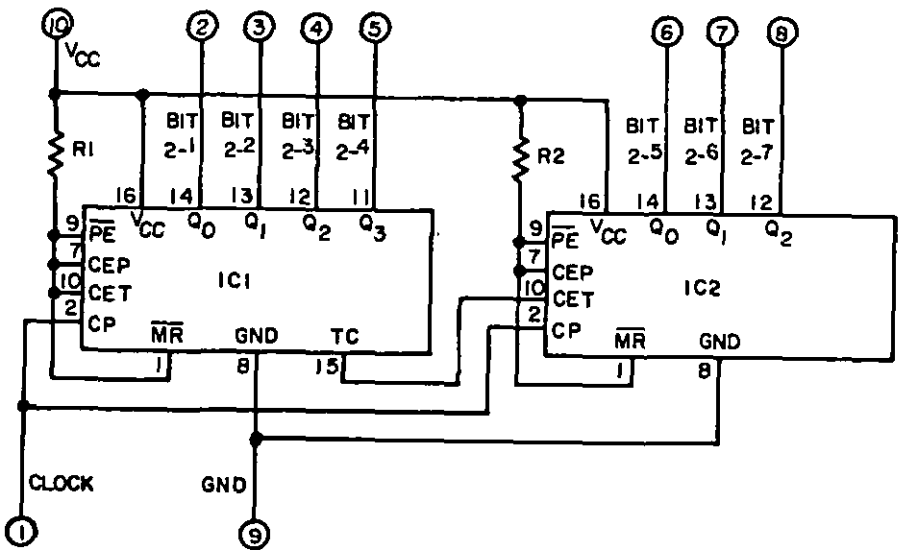
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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 50Ω ±5%, 1 W.
3. R3 = 1 kΩ ±1%, 1/8 W.
4. A1 = μA741, or equivalent.
5. Item G03 setup:
   Amplitude = 1 Vrms ±1%
   Frequency = 400 ±4 Hz.
6. Item J01 setup:
   Amplitude = 4 V ±10%
   Pulse width = 200 ±20 ns
   PRR = 1.0 ±0.1 MHz
   $t_{TLH} = t_{THL}$ = 25 ±20 ns.

FIGURE 3. Operating temperature and life test circuit.

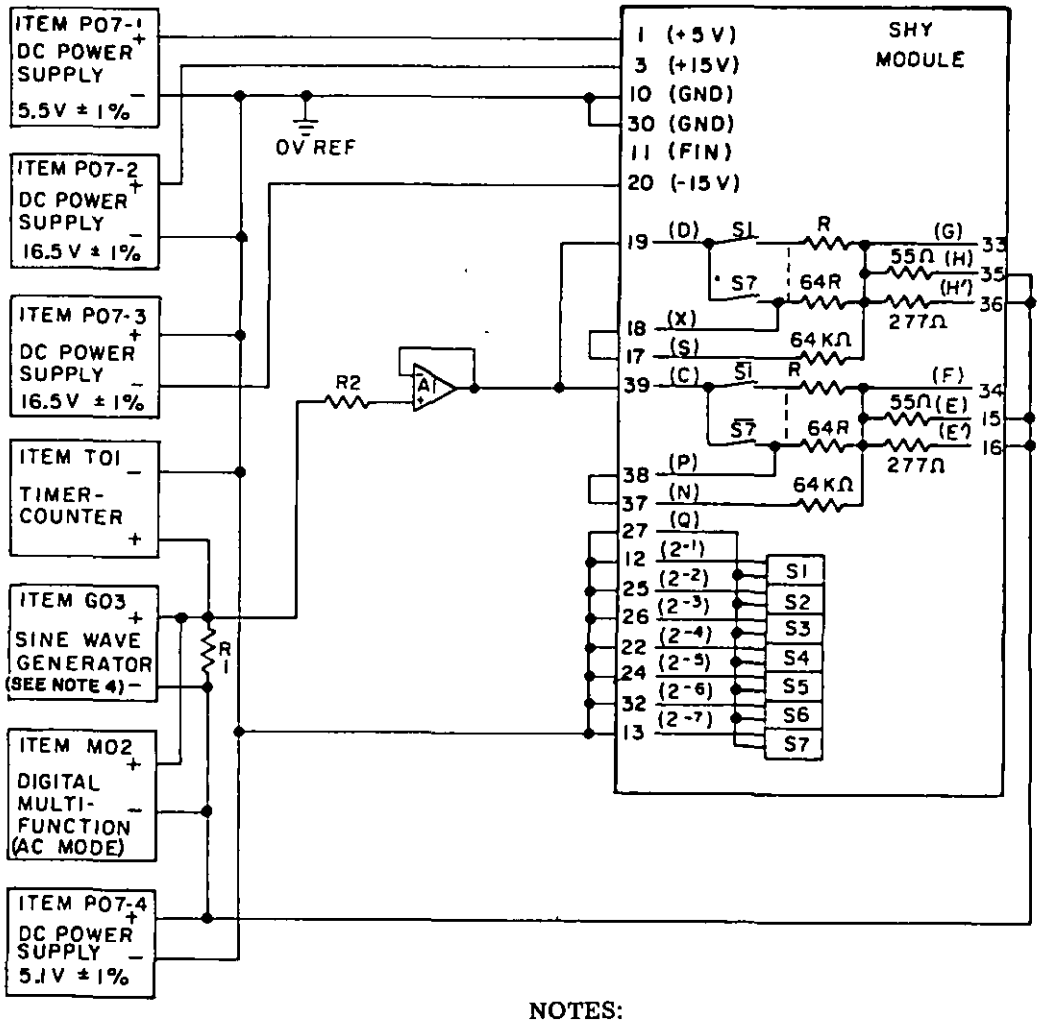
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NOTES:
1. R1 = R2 = 2.2 kΩ ±5%, 1/8 W.
2. IC1 = IC2 = 9316, or equivalent.
3. V_{CC} = 5.0 ±0.5 V.

FIGURE 4. Binary counter.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 1 kΩ ±1%, 1/8 W.
3. A1 = μA741, or equivalent.
4. Item G03 setup:
   Amplitude = 1.0 ±0.1 Vrms
   Frequency = 400 ±4 Hz.
5. Repeat test with pin 27 open.

FIGURE 5. Component temperature test circuit.

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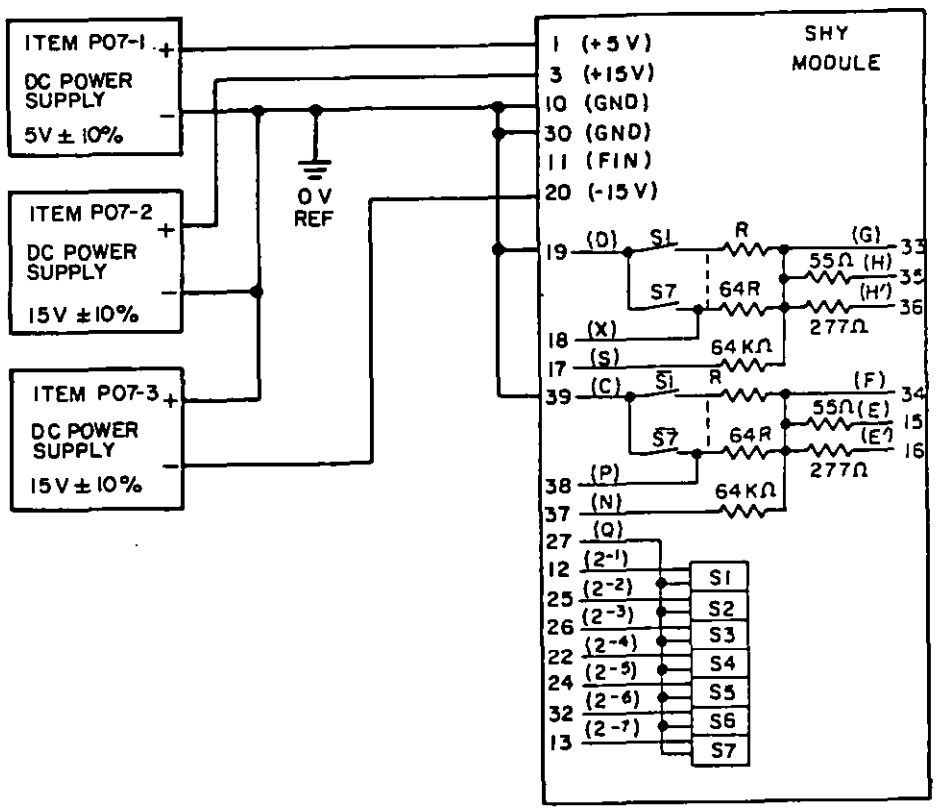


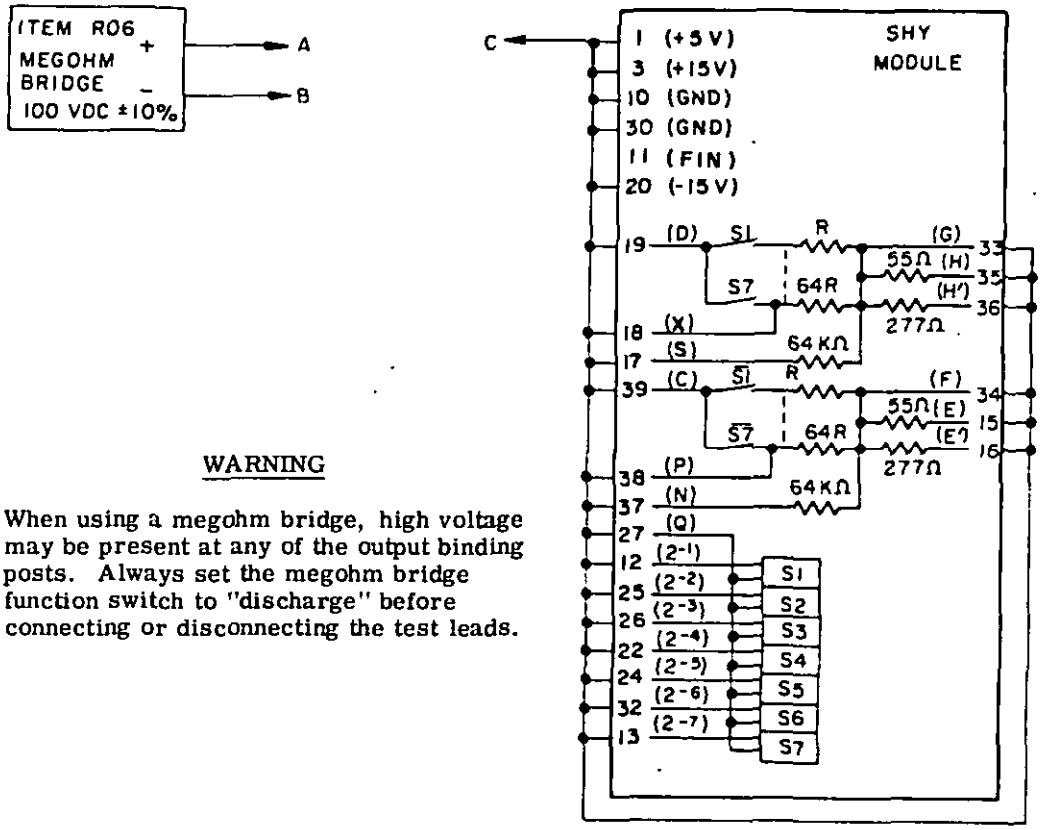
FIGURE 6. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, module pins to fin pin	C, 2, 4, 5, 6, 7, 8, 9, 14, 21, 23, 28, 29, 31, 40	11	10	∞	MΩ	7
A2	Isolation, unused to used pins	2, 4, 5, 6, 7, 8, 9, 14, 21, 23, 28, 29, 31, 40	C				
A3	Isolation, unused to unused pins	2, 4, 5, 6, 7, 8, 9, 14, 21, 23, 28, 29, 31	40				
A4		2, 4, 5, 6, 7, 8, 9, 14, 21, 23, 28, 29	31				
A5		2, 4, 5, 6, 7, 8, 9, 14, 21, 23, 28	29				
A6		2, 4, 5, 6, 7, 8, 9, 14, 21, 23	28				
A7		2, 4, 5, 6, 7, 8, 9, 14, 21	23				
A8		2, 4, 5, 6, 7, 8, 9, 14	21				
A9		2, 4, 5, 6, 7, 8, 9	14				
A10		2, 4, 5, 6, 7, 8	9				
A11		2, 4, 5, 6, 7	8				
A12		2, 4, 5, 6	7				
A13		2, 4, 5	6				
A14		2, 4	5				
A15	Isolation, unused to unused pins	2	4	10	∞	MΩ	7

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

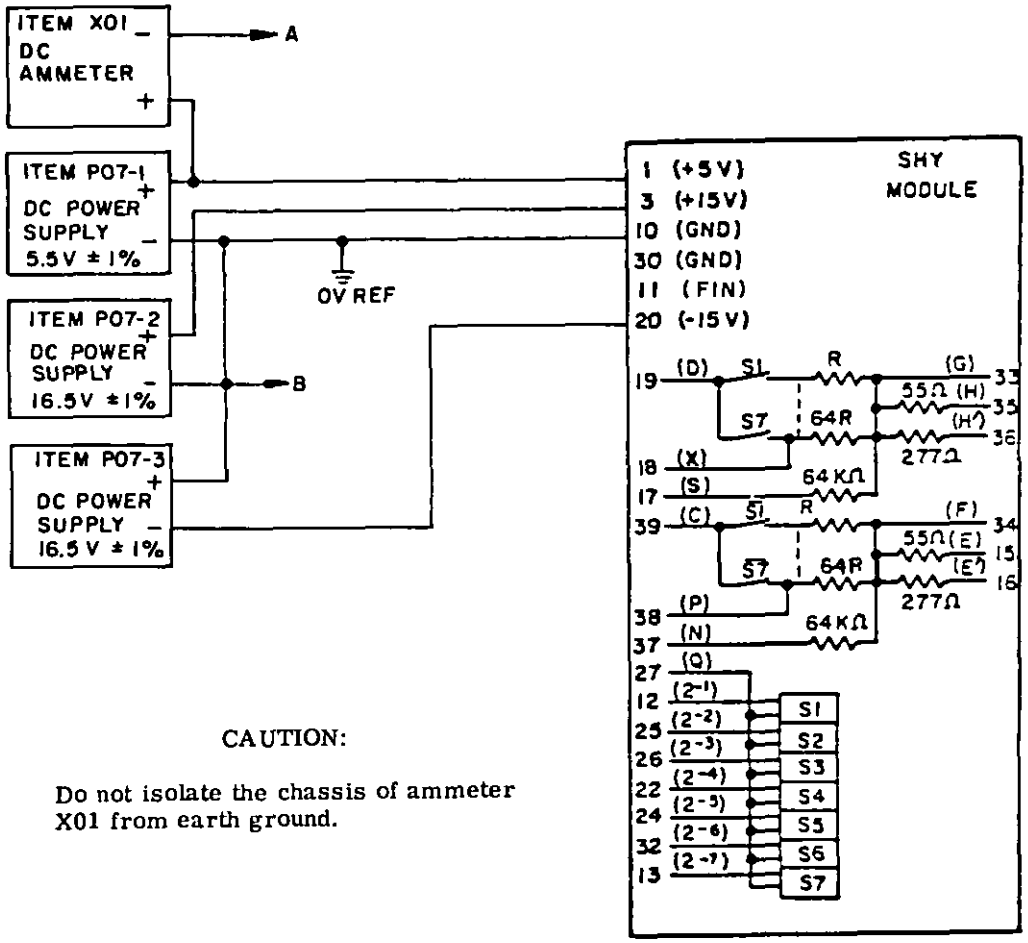
FIGURE 7. Isolation test circuit for table II.

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TABLE III. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	Q	Input breakdown current	27	12, 13, 22, 24, 25, 26, 32	0	1	mA	8
B2	2^{-1}		12	13, 22, 24, 25, 26, 27, 32				
B3	2^{-2}		25	12, 13, 22, 24, 26, 27, 32				
B4	2^{-3}		26	12, 13, 22, 24, 25, 27, 32				
B5	2^{-4}		22	12, 13, 24, 25, 26, 27, 32				
B6	2^{-5}		24	12, 13, 22, 25, 26, 27, 32				
B7	2^{-6}		32	12, 13, 22, 24, 25, 26, 27				
B8	2^{-7}	Input breakdown current	13	12, 22, 24, 25, 26, 27, 32	0	1	mA	8

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 8. Input breakdown current test circuit for table III.

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TABLE IV. Hot socket test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Hot socket	27	Open	26.262	26.328	Degrees	9
C2	↓	Open	27	63.672	63.738	↓	↓
C3		27	Open	26.262	26.328		
C4		Open	27	63.672	63.738		
C5		27	Open	26.262	26.328		
C6	Hot socket	Open	27	63.672	63.738	Degrees	9

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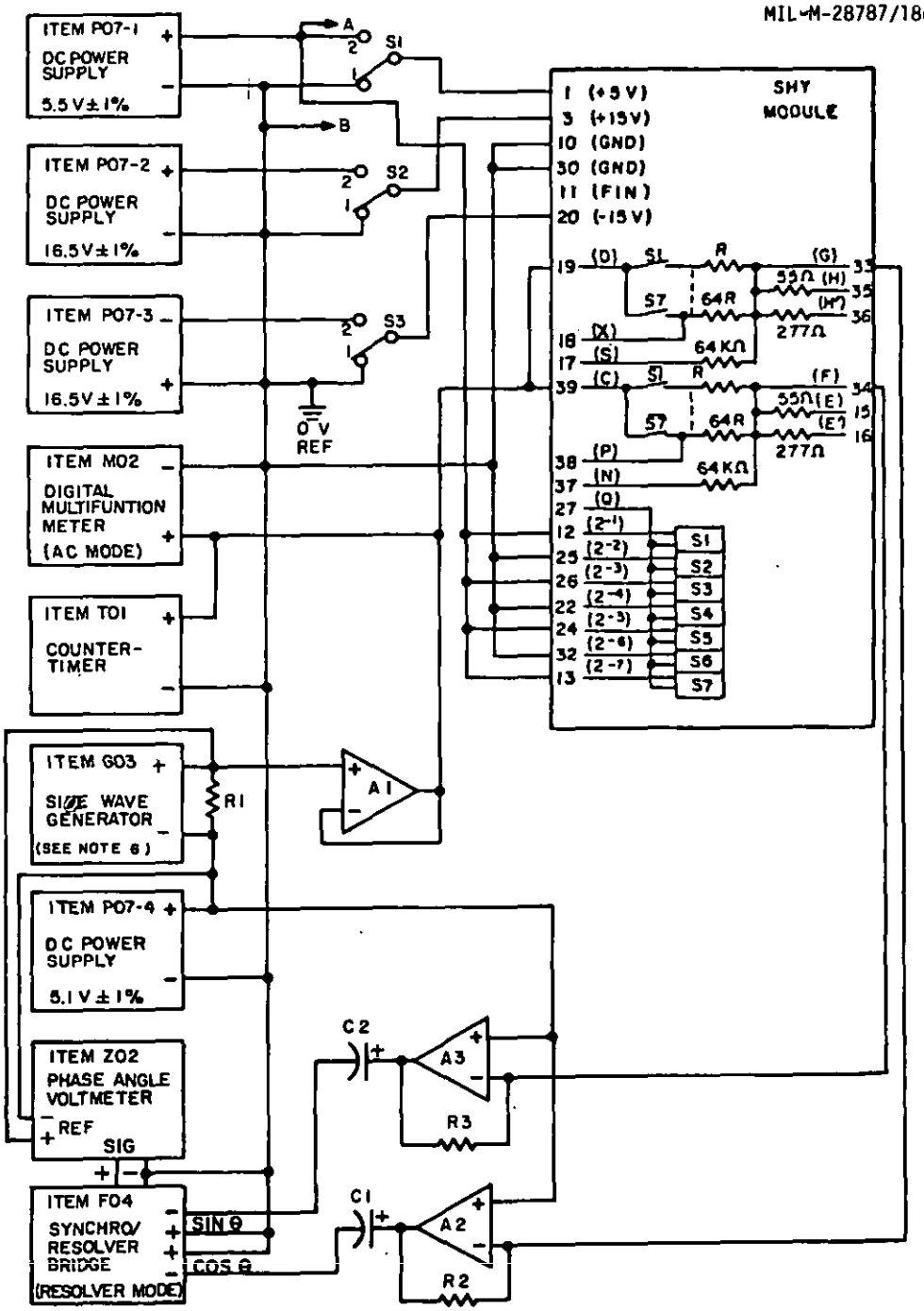


FIGURE 9. Hot socket test circuit for table IV.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 500Ω ±0.01% (matched to ±0.002%), 1/8 W.
3. C1 = C2 = 100 μF ±5%, 20 WVdc, electrolytic.
4. A1-A3 = μA741, or equivalent.
5. S1-S3 = SPDT switch.
6. Item G03 setup (verified at pins 19 and 39):
   Amplitude = 3.182 ±0.01 Vrms
   Frequency = 400 Hz ±1%.
7. Switching sequence:

   (a) For test condition C1, set switches S1, S2, and S3 to position 1, then operate switch S1 to position 2, S2 to position 2, and S3 to position 2 in this specified sequence.
   (b) Test condition C2 must be performed following test condition C1 without power disruption.
   (c) For test condition C3, set switches S1, S2, and S3 to position 1, then operate switch S2 to position 2, S3 to position 2, and S1 to position 2 in this specified sequence.
   (d) Test condition C4 must be performed following test condition C3 without power disruption.
   (e) For test condition C5, set switches S1, S2, and S3 to position 1, then operate switch S3 to position 2, S1 to position 2, and S2 to position 2 in this specified sequence.
   (f) Test condition C6 must be performed following test condition C5 without power disruption.

8. This test verifies an angular accuracy of ±0.033 degrees.

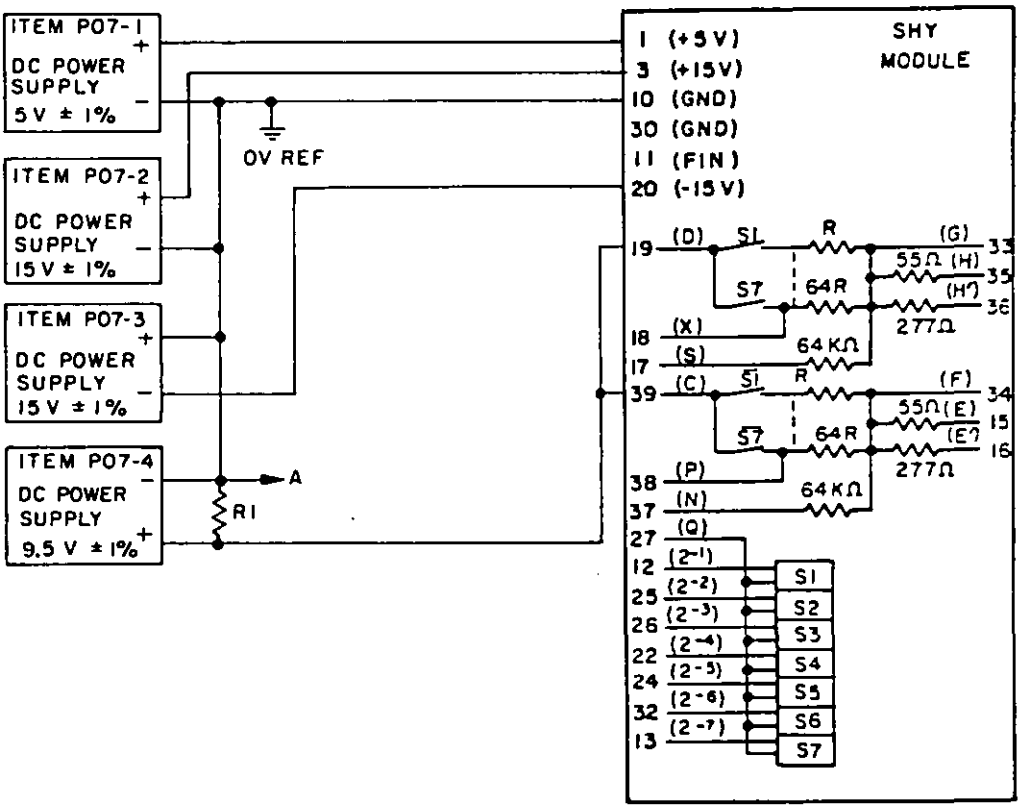
FIGURE 9. Hot socket test circuit for table IV - Continued.

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TABLE V. Switch check test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
D1	Switch check	12, 13, 22, 24, 25, 26	See figure 10, note 1			10
D2		12, 13, 22, 24, 25, 32				
D3		12, 13, 22, 24, 26, 32				
D4		12, 13, 22, 25, 26, 32				
D5		12, 13, 24, 25, 26, 32				
D6		12, 22, 24, 25, 26, 32				
D7		13, 22, 24, 25, 26, 32				
D8		12, 13, 22, 24, 25, 26, 27				
D9		12, 13, 22, 24, 25, 27, 32				
D10		12, 13, 22, 24, 26, 27, 32				
D11		12, 13, 22, 25, 26, 27, 32				
D12		12, 13, 24, 25, 26, 27, 32				
D13		12, 22, 24, 25, 26, 27, 32				
D14	Switch check	13, 22, 24, 25, 26, 27, 32	See figure 10, note 1			10

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NOTES:
1. This test is a design stress test. No measurements are made during test, but the module must meet all X type tests after subjection to test conditions D1-D14.
2. For test conditions D1-D14, point A shall be connected as specified in table V for a minimum of 5 seconds.
3. R1 = 50Ω ±5%, 2 W.

FIGURE 10. Switch check test circuit for table V.

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TABLE VI. Vs stress test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
E1	Vs stress	12,13,22,24,25,26, 27,32	359.967	0.033	Degrees	11
E2	Vs stress	12,13,22,24,25,26, 32	89.967	90.033	Degrees	11

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TABLE VII. 400 Hz angular accuracy test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
F1	400 Hz angular accuracy	12,13,22,24,25,26,27,32	359.967	0.033	Degrees	11
F2		12,22,24,25,26,27,32	0.4220	0.4880		
F3		12,22,24,25,26,27	1.3530	1.4190		
F4		12,22,25,26,27	3.3050	3.3710		
F5		12,25,26,27	7.5950	7.6610		
F6		12,25,27	17.8630	17.9290		
F7		12,27	44.5150	44.5810		
F8		27	89.967	90.033		
F9		13,27	89.5120	89.5780		
F10		13,27,32	88.5810	88.6470		
F11		13,24,27,32	86.6290	86.6950		
F12		13,22,24,27,32	82.3390	82.4050		
F13		13,22,24,26,27,32	72.0710	72.1370		
F14		13,22,24,25,26,27,32	45.4180	45.4840		
F15		22,25,27,32	63.6720	63.7380		
F16	400 Hz angular accuracy	22,25,32	26.2620	26.3280	Degrees	11

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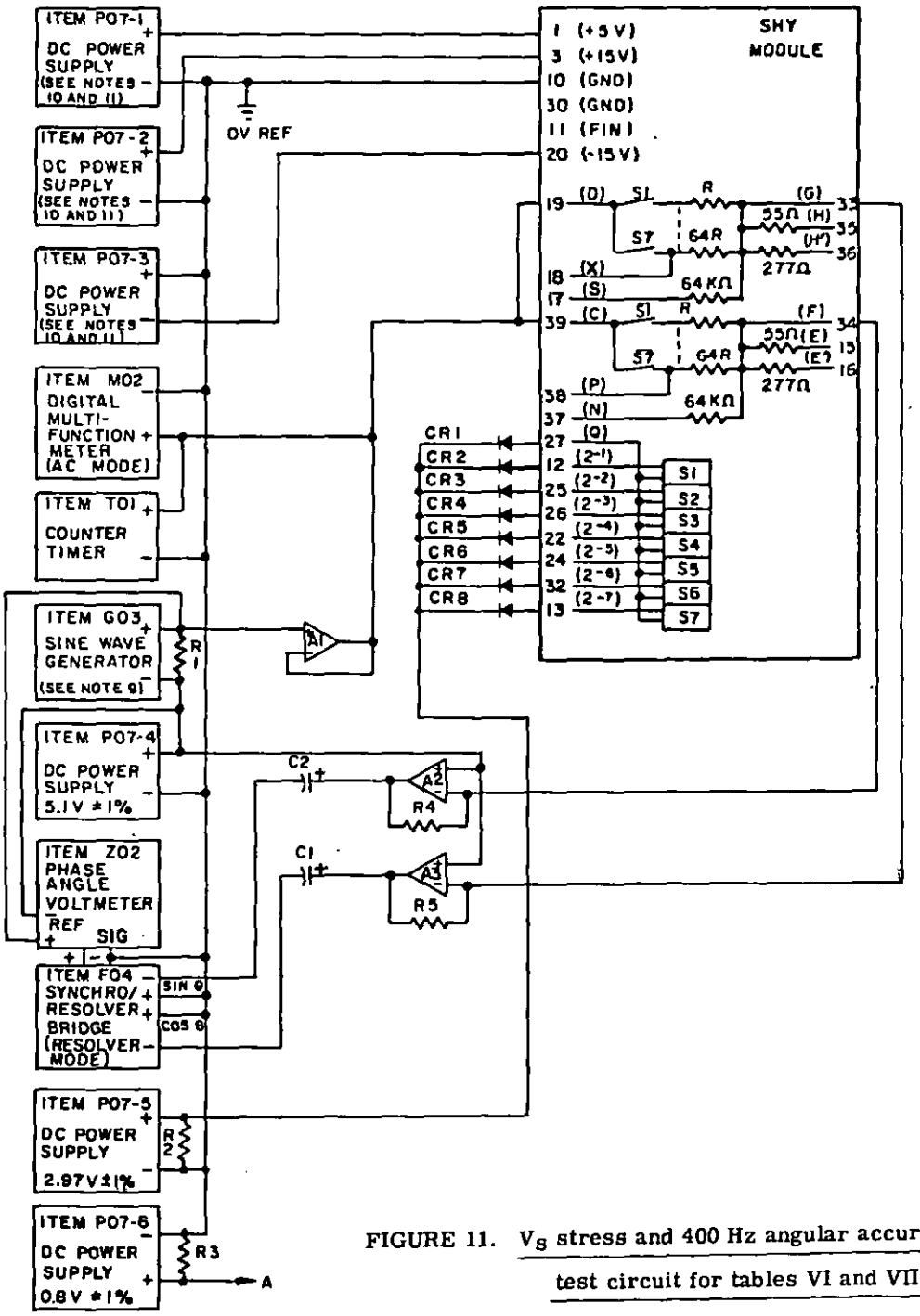


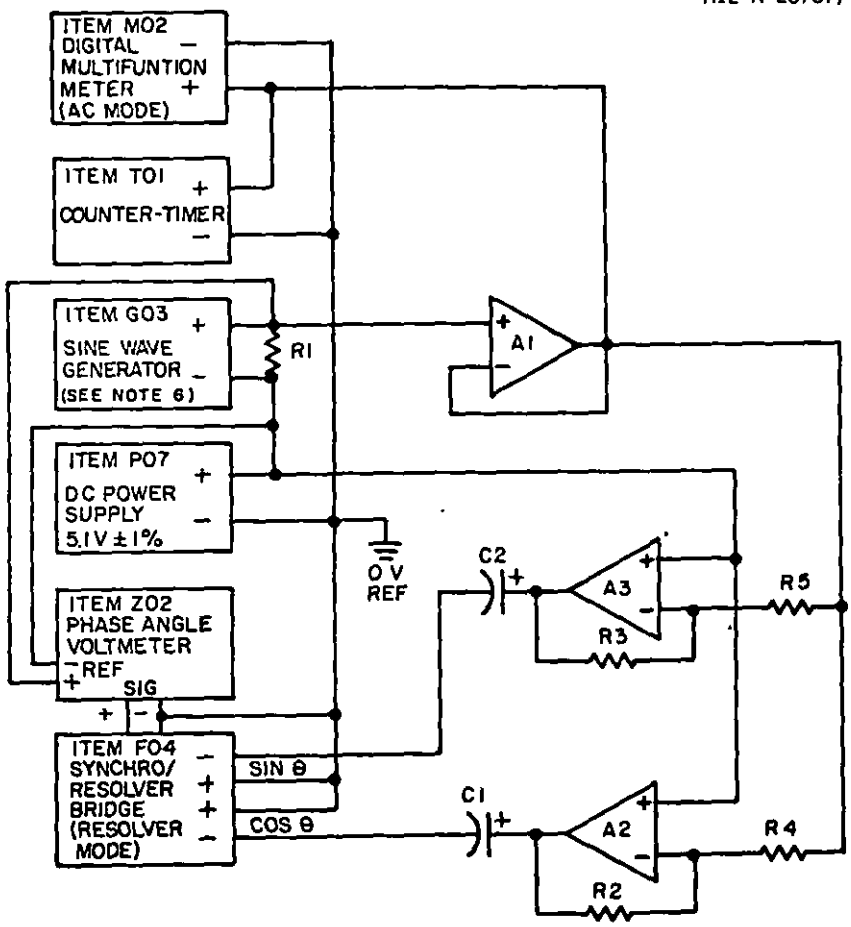
FIGURE 11. V_S stress and 400 Hz angular accuracy test circuit for tables VI and VII.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 50Ω ±5%, 1 W.
3. R3 = 5Ω ±5%, 1/2 W.
4. R4 = R5 = 500Ω ±0.01% (matched to ±0.002%), 1/8 W.
5. C1 = C2 = 100 μF ±5%, 20 WVdc, electrolytic.
6. CR1-CR8 = 1N4150, or equivalent.
7. A1-A3 = μA741, or equivalent.
8. Amplifiers A2 and A3 are to be matched per figure 12.
9. Item G03 setup (verified at pins 19 and 39):
   Amplitude = 3.18 ±0.01 Vrms
   Frequency = 400 ±4 Hz.
10. The Vs stress test verifies correct module operation after the module power inputs were stressed for a minimum of 5 seconds and a maximum of 30 seconds.
    (a) Initially adjust power supply P07-1 for 7 V ±1% and power supplies P07-2 and P07-3 for 18 V ±1%.
    (b) Without disconnecting the test circuit, decrease the voltage of P07-1 to 5 V ±1% and power supplies P07-2 and P07-3 to 15 V ±1%.
    (c) For test condition E1, null bridge F04 and record reading.
    (d) Without power interruption, make E2 measurement.
11. Test condition F verifies an angular accuracy of ±0.033 degrees.
    (a) Adjust power supply P07-1 for 4.5 V ±1% and power supplies P07-2 and P07-3 for 13.5 V ±1%.
    (b) Null bridge F04 and record reading.

FIGURE 11. V_S stress and 400 Hz angular accuracy test circuit for tables VI and VII - Continued.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 500Ω ±0.01% (matched to ±0.002%), 1/8 W.
3. R4 = R5 = 1 kΩ ±1% (matched to ±0.005%), 1/8 W.
4. C1 = C2 = 100 μF ±5%, 20 WVdc, electrolytic.
5. A1-A3 = μA741, or equivalent.
6. Item G03 setup (verified at output of A1):
   Amplitude = 3.2 ±0.01 Vrms.
   Frequency = 400 ±4 Hz.
7. The null for bridge F04 shall be between 44.9970 and 45.0030 degrees.

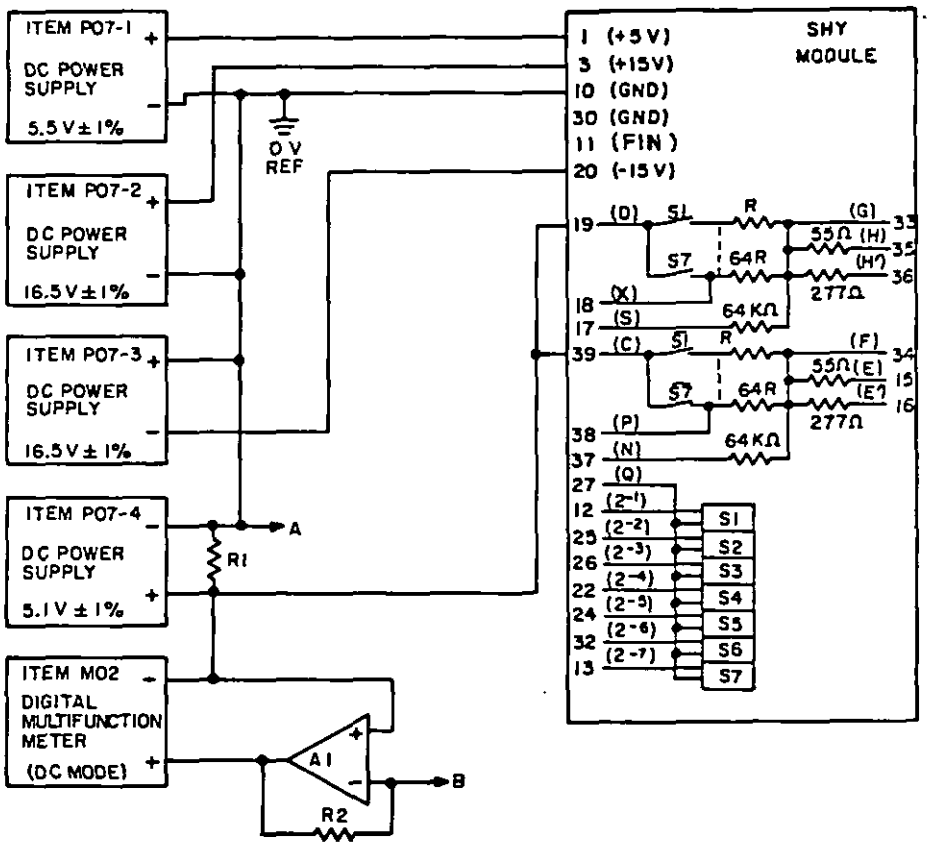
FIGURE 12. Test circuit to match A2 and A3.

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TABLE VIII. DC integrity and leakage check test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	DC integrity and leakage check	12	34	1.0	-96.0	mV	13
G2		25			-48.0		
G3		26			-24.0		
G4		22			-12.0		
G5		24			- 6.0		
G6		32			- 3.0		
G7		13	34		- 1.5		
G8		13, 22, 24, 25, 26, 32	33		-96.0		
G9		12, 13, 22, 24, 26, 32			-48.0		
G10		12, 13, 22, 24, 25, 32			-24.0		
G11		12, 13, 24, 25, 26, 32			-12.0		
G12		12, 13, 22, 25, 26, 32			- 6.0		
G13		12, 13, 22, 24, 25, 26			- 3.0		
G14		12, 22, 24, 25, 26, 32			- 1.5		
G15		12, 13, 22, 24, 25, 26, 32	33		- 1.5		
G16		Open	34		- 1.5		
G17		12, 13, 22, 24, 25, 26, 32	34		-192.0		
G18	DC integrity and leakage check	Open	33	1.0	-192.0	mV	13

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 64.9 kΩ ±1%, 1/8 W.
3. A1 = AD233K, or equivalent.
4. DC offset of A1 shall be less than 0.1 mVdc.
5. This test verifies that each switch has less than 1.5 mV of dc offset.
6. Record measurement as read on meter M02.

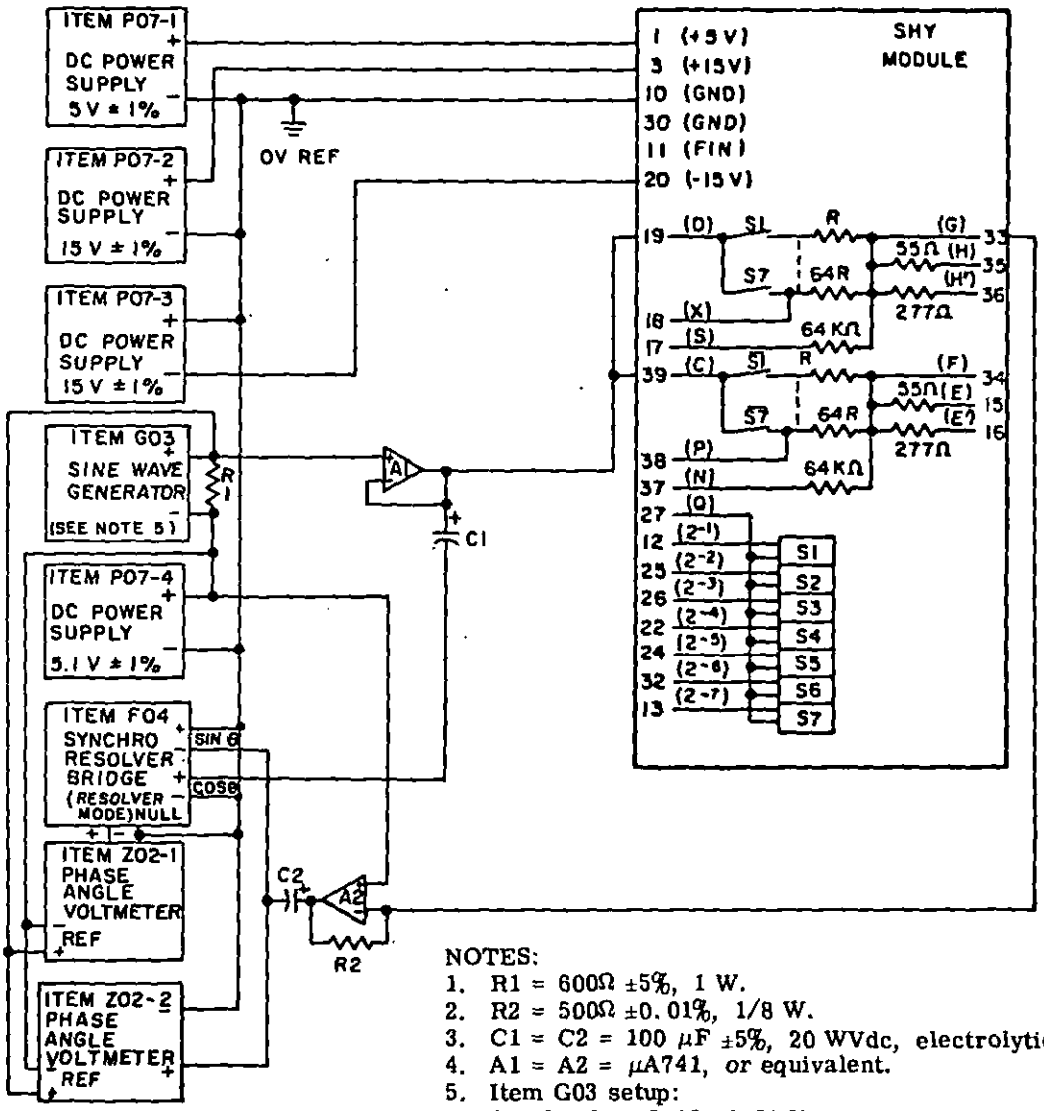
FIGURE 13. DC integrity and leakage check test circuit for table VIII.

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TABLE IX. Absolute transformation ratio check and phase shift test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
H1	Absolute transformation ratio check	44.6324	44.9189	Degrees	14
H2	Phase shift	179.5	180.5	Degrees	14

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 500Ω ±0.01%, 1/8 W.
3. C1 = C2 = 100 μF ±5%, 20 WVdc, electrolytic.
4. A1 = A2 = μA741, or equivalent.
5. Item G03 setup:
   Amplitude = 3.18 ±0.01 Vrms
   Frequency = 400 ±4 Hz.
6. For test condition H1, read bridge F04.
7. For test condition H2, read voltmeter Z02-2.

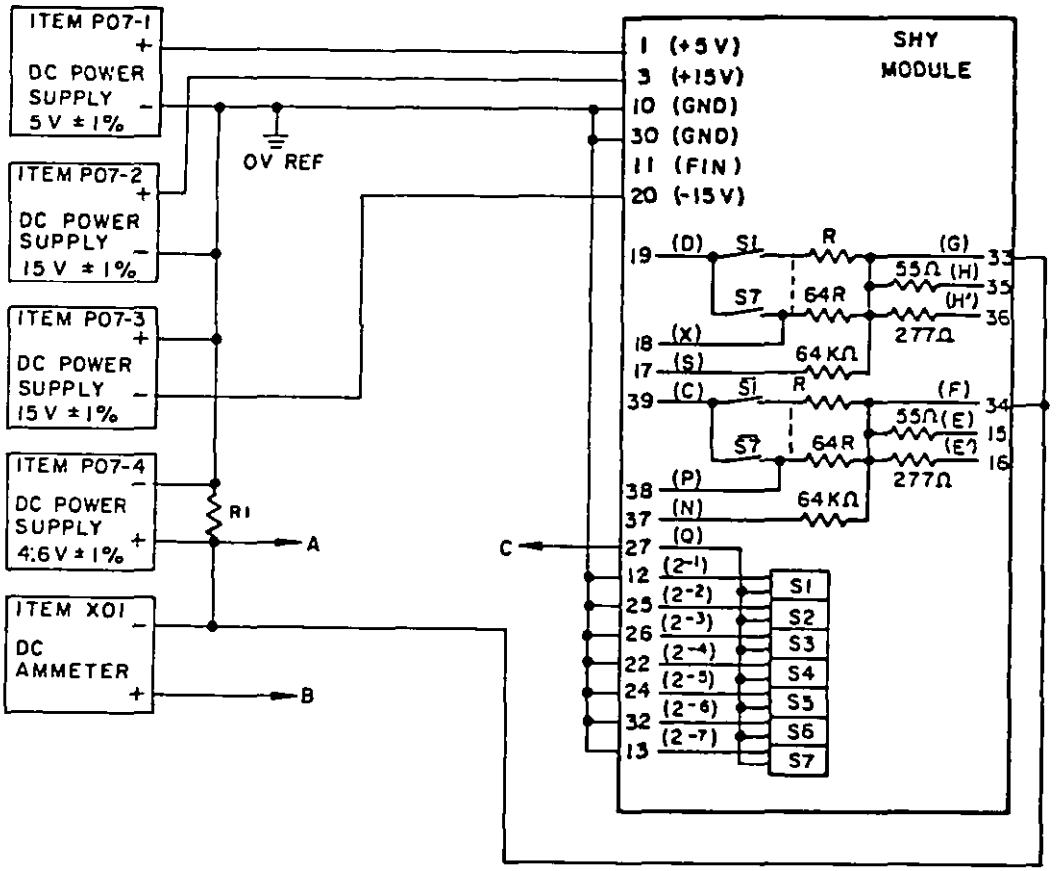
FIGURE 14. Absolute transformation ratio check and phase shift test circuit for table IX.

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TABLE X. Input current (analog) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Input current (analog), V_I = 4.6 V	19	39	10	0.0	4.8	mA	15
J2	↓	19	39	1	↓	↓	↓	↓
J3	↓	39	19	1	↓	↓	↓	↓
J4	Input current (analog) V_I = 4.6 V	39	19	10	0.0	4.8	mA	15

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: R1 = 50Ω ±5%, 1 W.

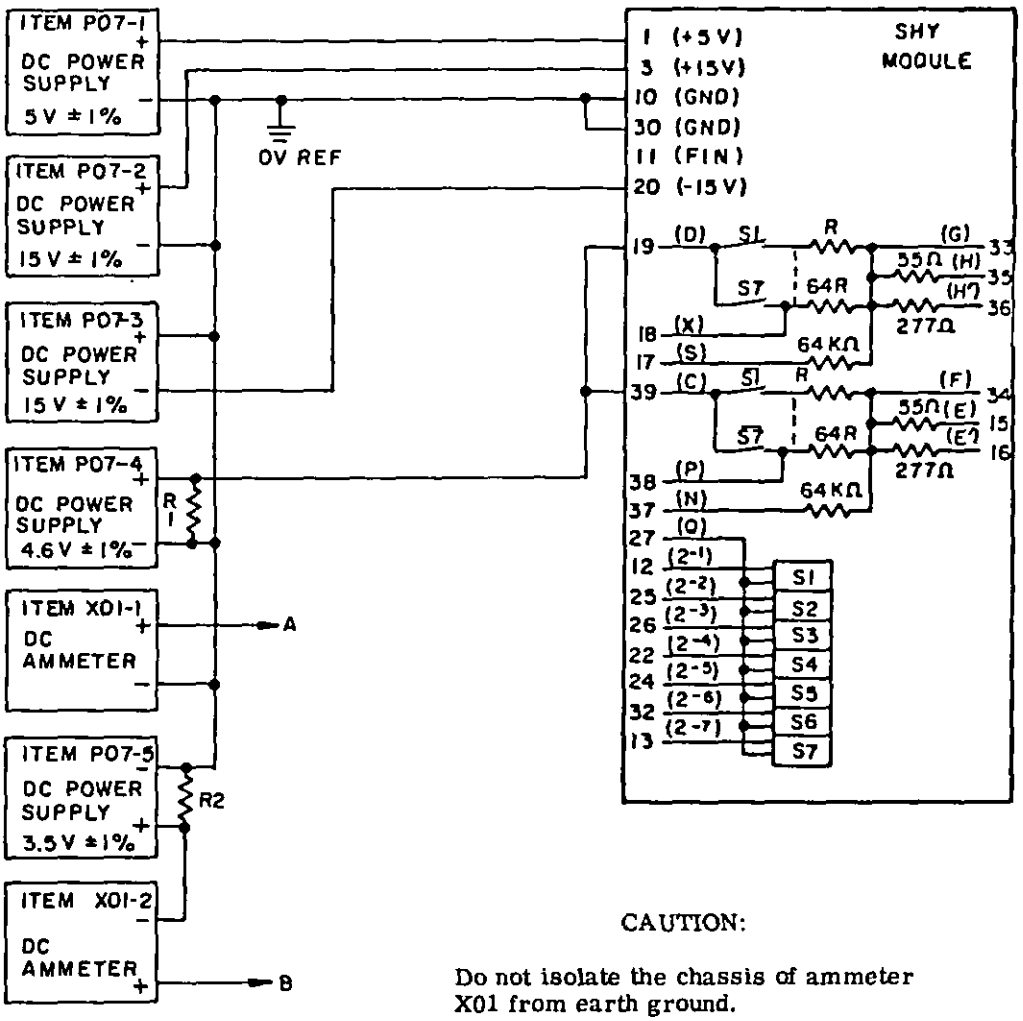
FIGURE 15. Input current (analog) test circuit for table X.

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TABLE XI. Input current (digital) test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K1	2^{-1}	Input current (digital), low level	12	Open	0.9	1.4	mA	16
K2	2^{-2}		25					
K3	2^{-3}		26					
K4	2^{-4}		22					
K5	2^{-5}		24					
K6	2^{-6}		32					
K7	2^{-7}		13					
K8	Q	Input current (digital), low level	27	Open	0.9	1.4	mA	
K9	2^{-1}	Input current (digital), high level	Open	12	275	440	µA	
K10	2^{-2}			25				
K11	2^{-3}			26				
K12	2^{-4}			22				
K13	2^{-5}			24				
K14	2^{-6}			32				
K15	2^{-7}			13				
K16	Q	Input current (digital), high level	Open	27	275	440	µA	16

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Since the ammeters are not used simultaneously, only one dc ammeter, X01, is required.
3. With point "B" disconnected, ammeter X01-2 should indicate zero current on the 1 μA range.

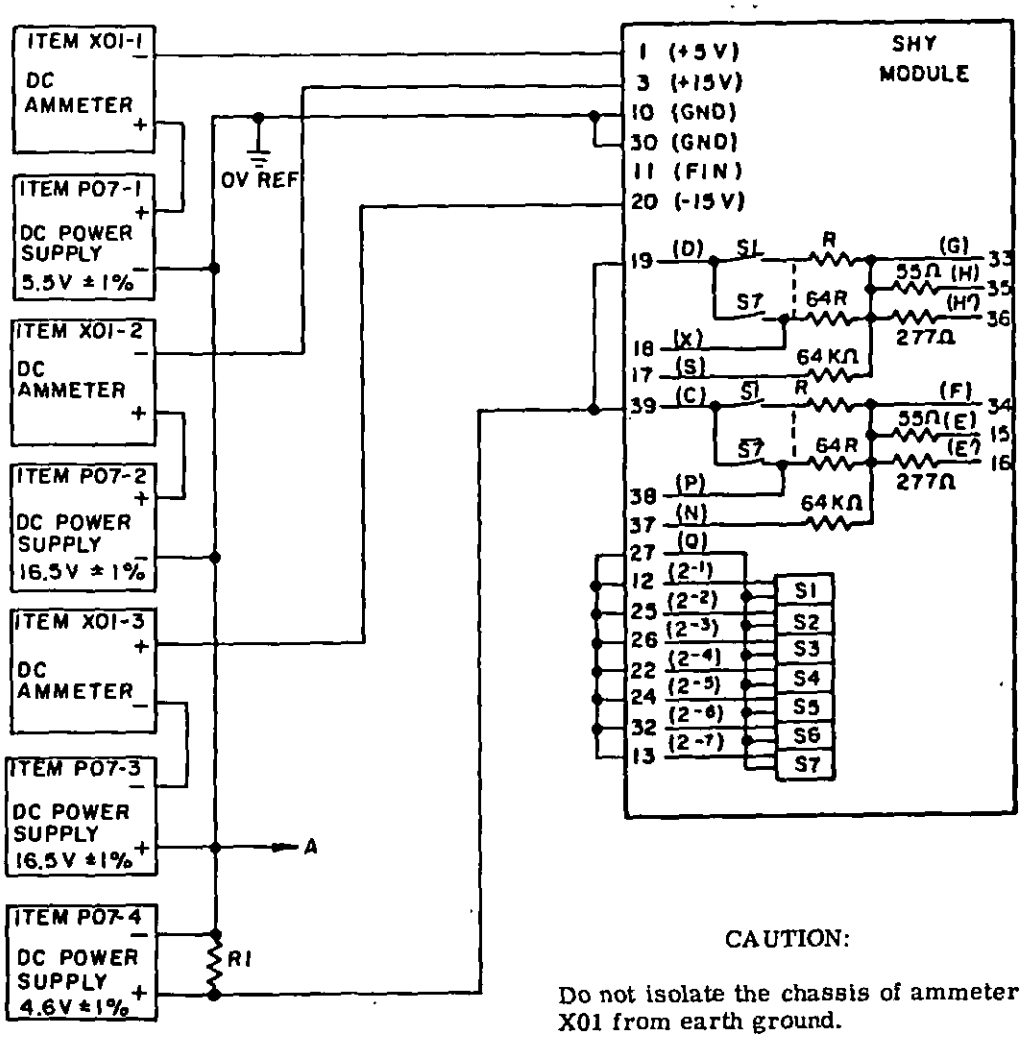
FIGURE 16. Input current (digital) test circuit for table XI.

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TABLE XII. Power supply current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Power supply current, 5.5 V	Open	0	1	mA	17
L2	Power supply current, + 16.5 V	Open	↓	20	↓	↓
L3	Power supply current, - 16.5 V	Open	↓	20	↓	↓
L4	Power supply current, 5.5 V	27	0	30	mA	17

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Since the ammeters are not used simultaneously, only one dc ammeter, X01, is required.

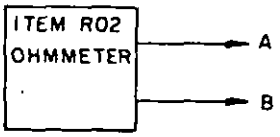
FIGURE 17. Power supply current test for table XII.

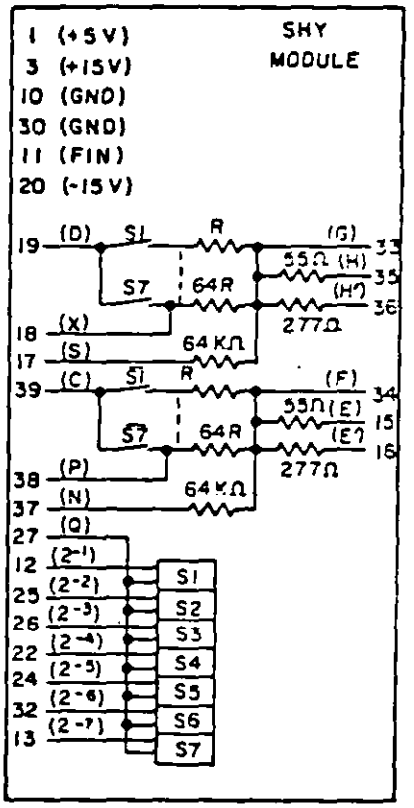
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TABLE XIII. Scaling resistor network test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Scaling resistor network	18	33	63.5	64.5	kΩ	18
M2	↓	17	33	↓	↓	↓	↓
M3		37	34				
M4		38	34	63.5	64.5	kΩ	
M5		35	33	54.45	55.55	Ω	
M6		15	34	54.45	55.55	↓	
M7		36	33	274.2	279.8		
M8	Scaling resistor network	16	34	274.2	279.8		
M9	Calculation (M1 -M3)	—	--	-500	500		
M10	Calculation (M1 -M4)	—	--	↓	↓		
M11	Calculation (M2 -M3)	—	--				
M12	Calculation (M2 -M4)	—	--	-500	500		
M13	Calculation (M5 -M6)	—	--	-0.055	0.055	↓	↓
M14	Calculation (M7 -M8)	—	--	-0.278	0.278	Ω	18

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NOTES:

1. For test conditions M9-M12:

$$\frac{|Mi - Mj|}{64{,}000} \leq 0.0078 \longrightarrow \begin{array}{l} |M1 - M3| \leq 500; \quad |M1 - M4| \leq 500; \\ |M2 - M3| \leq 500; \quad |M2 - M4| \leq 500. \end{array}$$

2. For test condition M13:

$$\frac{|M5 - M6|}{55} \leq 0.001 \longrightarrow |M5 - M6| \leq 0.055.$$

3. For test condition M14:

$$\frac{|M7 - M8|}{277.55} \leq 0.001 \longrightarrow |M7 - M8| \leq 0.278.$$

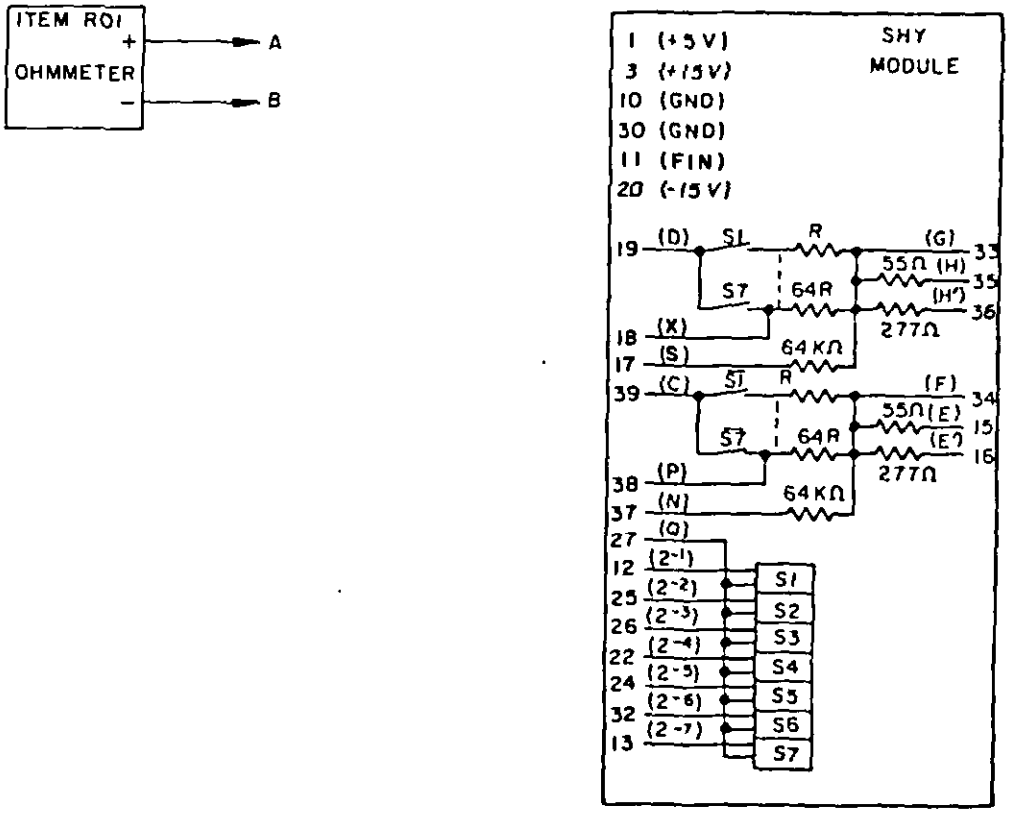
FIGURE 18. Scaling resistor network test for table XIII.

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TABLE XIV. Continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Continuity, ground	10	30	0.0	0.1	Ω	19
N2	Continuity, ground	10	30	0	1	Ω	19
N3	Continuity, fin	11	Fin	0	1	Ω	19

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NOTE: For test condition N3, penetrate the anodized finish on the module fin with a sharply pointed probe connected to the negative terminal.

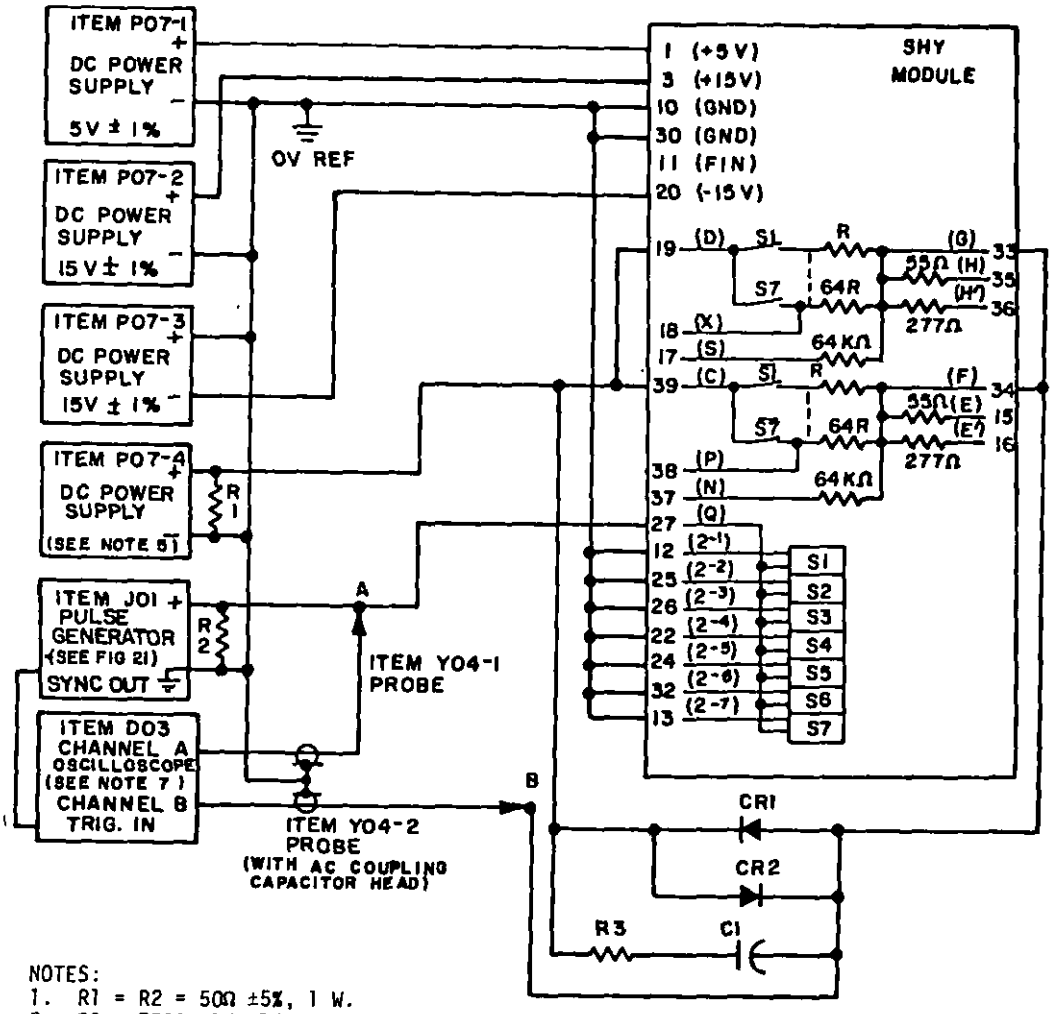
FIGURE 19. Continuity test circuit for table XIV.

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TABLE XV. Settling time test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
P1	Settling time (t_{S1})	0	4	µs	20, 21
P2	Settling time (t_{S2})	↓	↓	↓	↓
P3	Settling time (t_{S1})	↓	↓	↓	↓
P4	Settling time (t_{S2})	0	4	µs	20, 21

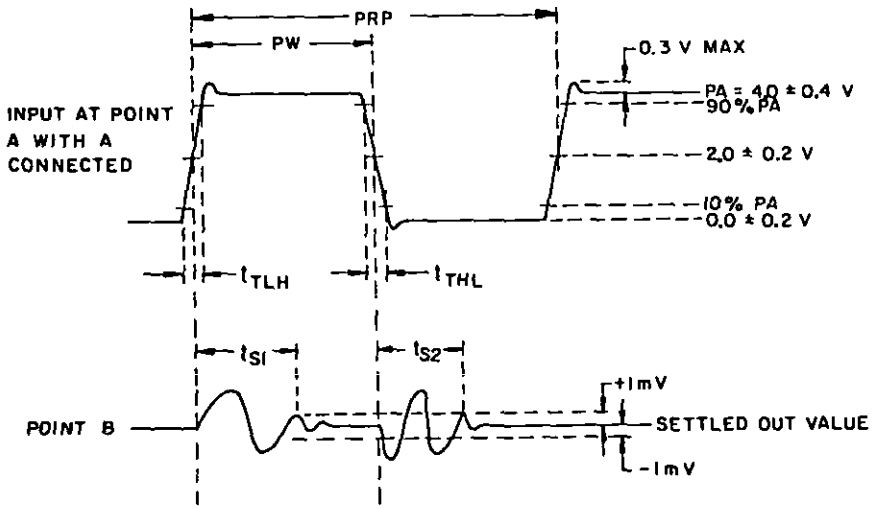
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. R3 = 750Ω ±1%, 1/8 W.
3. C1 = 220 pF ±5%, 20 WVdc.
4. CR1 = CR2 = 1N4148, or equivalent.
5. For test conditions P1 and P2, adjust power supply P07-4 for 6.0 V ±1% and for test conditions P3 and P4, adjust power supply P07-4 for 1.1 V ±1%.
6. Oscilloscope D03 shall be ac coupled so that channel B can be amplified for maximum amplitude or a dc voltage can be subtracted by means of a differential comparator. Verify that channel B output amplitude stays within ±1 mV of its settled out value at 4.0 µs after the input pulse.
7. Use type W plug-in.

FIGURE 20. Settling time test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   (a) PW = 10.0 ±0.5 μs.
   (b) PRP = 20 ±1 μs.
   (c) $t_{TLH} = t_{THL}$ = 50 ±5 ns.
2. $t_{S1} = t_{S2} \leq 4.0$ μs.

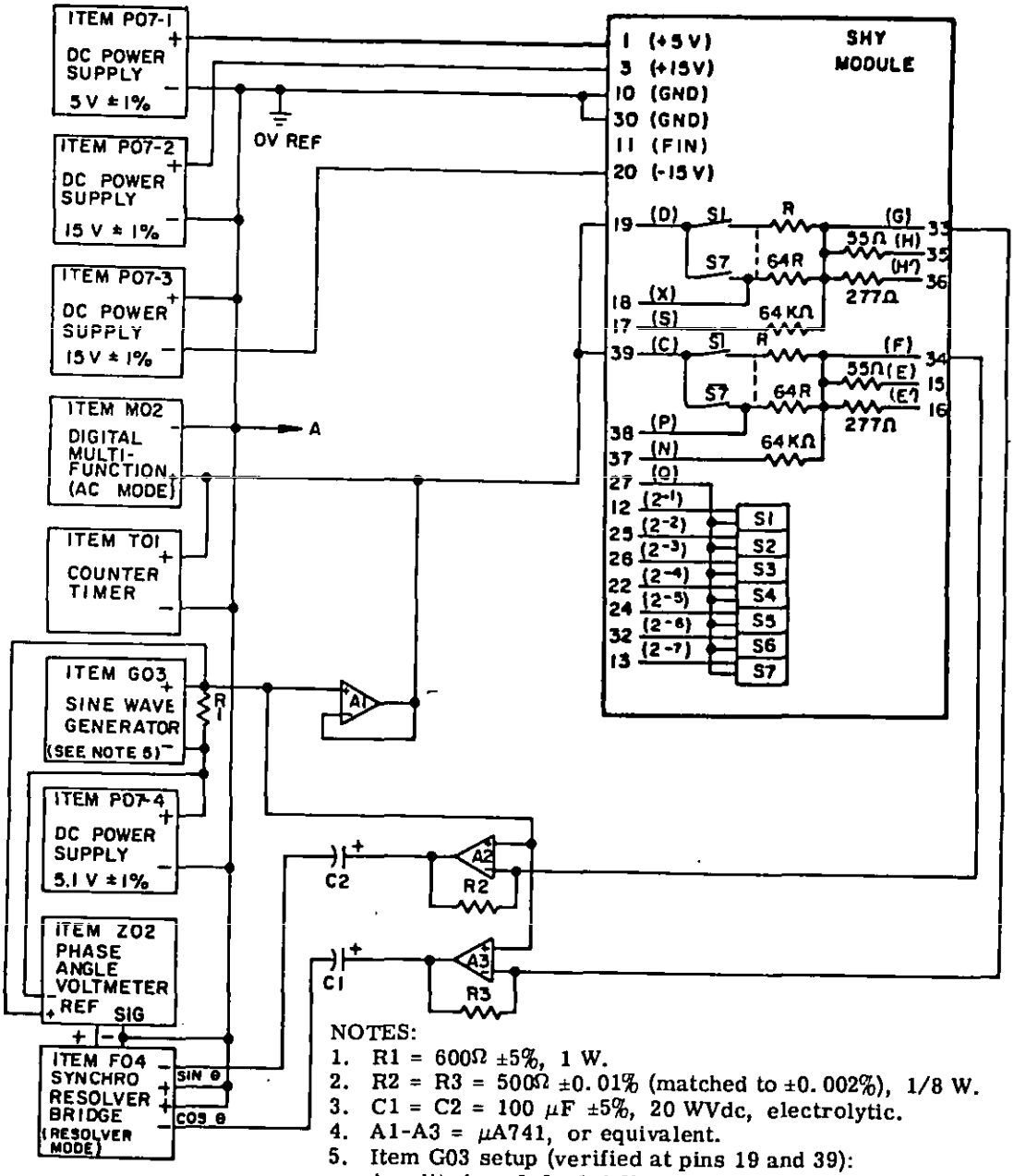
FIGURE 21. Settling time test circuit waveform for table XV.

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TABLE XVI. 800 Hz angular accuracy test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
Q1	800 Hz angular accuracy	12, 13, 24, 26	63.672	63.738	Degrees	22
Q2	800 Hz angular accuracy	12, 13, 24, 26, 27	26.262	26.328	Degrees	22
Q3	800 Hz angular accuracy	22, 25, 27, 32	63.672	63.738	Degrees	22

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 500Ω ±0.01% (matched to ±0.002%), 1/8 W.
3. C1 = C2 = 100 μF ±5%, 20 WVdc, electrolytic.
4. A1-A3 = μA741, or equivalent.
5. Item G03 setup (verified at pins 19 and 39):
   Amplitude = 3.2 ±0.1 Vrms.
   Frequency = 800 ±8 Hz.

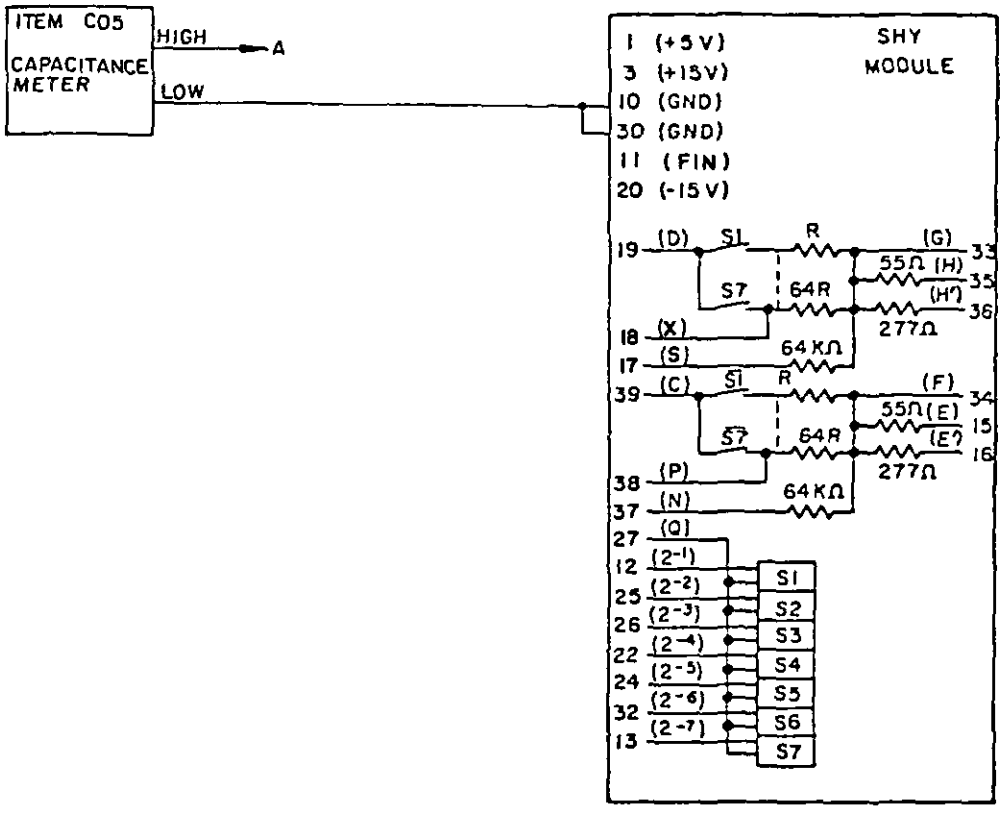
FIGURE 22. 800 Hz angular accuracy test circuit for table XVI.

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TABLE XVII. Input and output capacitance test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
R1	2^{-1}	Input capacitance	12	2	35	pF	23
R2	2^{-2}		25				
R3	2^{-3}		26				
R4	2^{-4}		22				
R5	2^{-5}		24				
R6	2^{-6}		32				
R7	2^{-7}		13				
R8	Q	Input capacitance	27	2	35		
R9	G	Output capacitance	33	50	100		
R10	F		34				
R11	H		35				
R12	H'		36				
R13	E		15				
R14	E'	Output capacitance	16	50	100	pF	23

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NOTES:
1. Null out effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and 30 and the pin under test. Stray capacitance added to other pins may affect the readings.

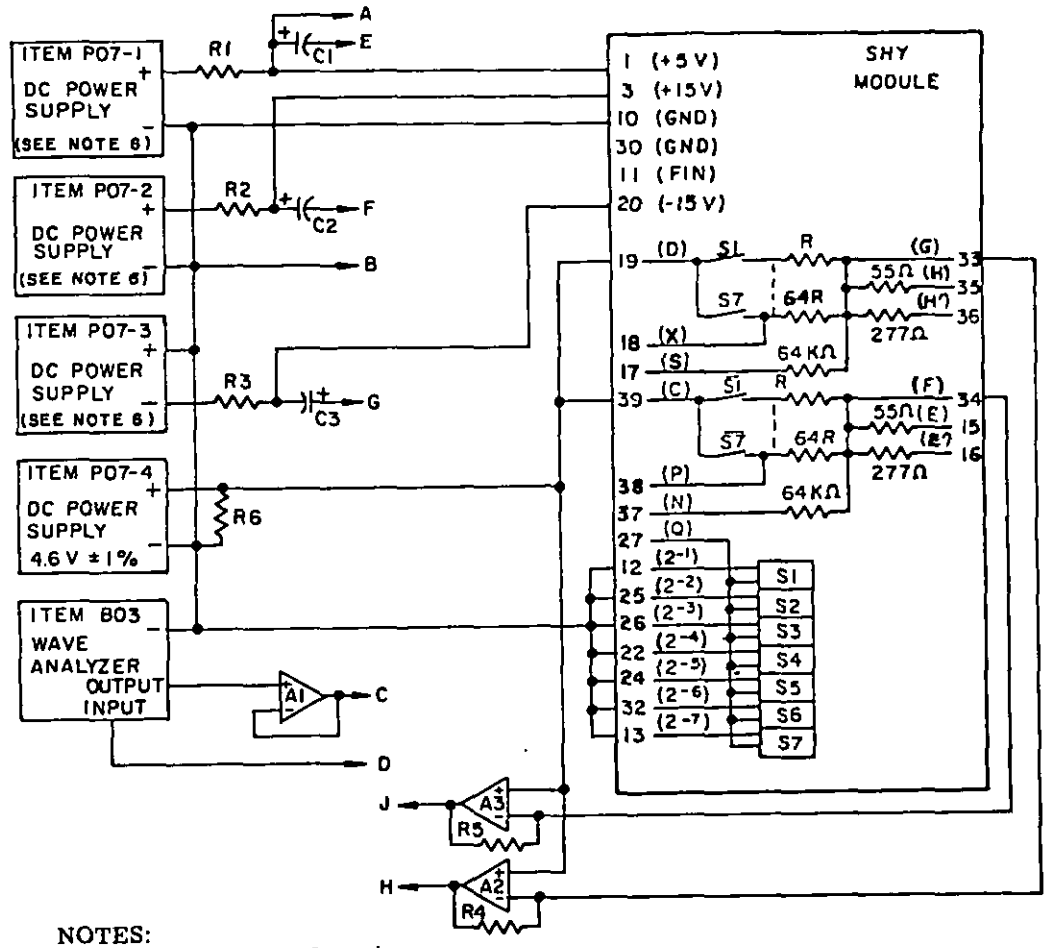
FIGURE 23. Input and output capacitance test for table XVII.

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TABLE XVIII. Power supply noise rejection test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	Power supply noise rejection, 5 V	27	Open	E	H	0	2	mVrms	24
S2	Power supply noise rejection, 5 V	Open	27	E	J				
S3	Power supply noise rejection, + 15 V	27	Open	F	H				
S4	Power supply noise rejection, + 15 V	Open	27	F	J				
S5	Power supply noise rejection, - 15 V	27	Open	G	H				
S6	Power supply noise rejection, - 15 V	Open	27	G	J		2		
S7	Power supply noise rejection, 5 V	27	Open	E	H		6		
S8	Power supply noise rejection, 5 V	Open	27	E	J				
S9	Power supply noise rejection, + 15 V	27	Open	F	H				
S10	Power supply noise rejection, + 15 V	Open	27	F	J				
S11	Power supply noise rejection, - 15 V	27	Open	G	H				
S12	Power supply noise rejection, - 15 V	Open	27	G	J	0	6	mVrms	24

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NOTES:

1. R1-R3 = 100Ω ±5%, 1/8 W.
2. R4 = R5 = 64 kΩ ±1% (matched to ±0.002%), 1/8 W.
3. R6 = 50Ω ±5%, 1 W.
4. C1-C3 = 100 μF ±5%, 20 WVdc, electrolytic.
5. A1-A3 = μA741, or equivalent.
6. Power supply setups:
   P07-1 = 5 V ±1% (verified at pin 1)
   P07-2 = 15 V ±1% (verified at pin 3)
   P07-3 = -15 V ±1% (verified at pin 20).
7. Item B03 output:

Amplitude	Frequency	Test condition
0.354 Vrms ±5%	50 Hz to 1 kHz	S1 and S2
0.707 Vrms ±5%	50 Hz to 1 kHz	S3 and S4
0.707 Vrms ±5%	50 Hz to 1 kHz	S5 and S6
0.354 Vrms ±5%	1 kHz to 10 kHz	S7 and S8
0.707 Vrms ±5%	1 kHz to 10 kHz	S9 thru S12

8. Item B03 bandwidth is to be 10 Hz.

FIGURE 24. Power supply noise rejection test circuit for table XVIII.

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